sanas South African National Accreditation System	**TECHNICAL REQUIREMENTS & VERTICAL CHECKLIST FOR PROFICIENCY TESTING SCHEMES** **ISO/IEC 17043:2023 Conformity Assessment – General Requirements for Proficiency Testing**	Organisation's SANAS No/s. Document Review

Date/s of evaluation	**March 2026**
Assessor/s & Observers	**Phuti Mathopa**
Name of Proficiency Testing Provider (PTP)	**Azuresky Pty Ltd**
Area / field of operation	**Coal analysis**
Laboratory Representative	

This report covers the following:

Document Review only	**X**	Implementation on Site Visit only		Document Review and Site Visit		Other	

REQUIREMENTS & COMMENTS.

Compliance = C, Non-compliance = NC. Where a clause is marked as NA, reason must be provided as to why it's not applicable

NB1: *References to ISO/IEC 17043:2023 are in italics. The order of assessment need not follow the order of the checklist. Assessors are expected to know & have the standard, this worksheet is designed as guidance to prompt detailed recording of the process.*

REFER TO ISO/IEC 17043:2023 FOR DETAIL AND FOR CLARIFICATION NOTES.

PART A: TECHNICAL REQUIREMENTS CHECKLIST				
CLAUSE	***ISO/IEC 17043:2023 REQUIREMENTS***	***CAB's COMMENTS*** *The CAB must provide information on **how** requirements have been addressed, documented and/or implemented. Make reference to policies / procedures, incl. clause numbers.*	**C/ NC**	***ASSESSOR's COMMENTS*** *Indicate **WHAT** has been checked and **HOW** requirements have been implemented.*
6	**Resource Requirement**			
6.1	***General***			
6.1.1	Does the PT provider have access to the personnel, facilities, equipment, systems and support services necessary to manage and perform its PT activities?			
6.1.2	Are the measurements or tests conducted under the responsibility of the PT provider, related to PT item characterization or for assessing homogeneity and stability, conducted in accordance with the relevant requirements of ISO/IEC 17025?			
6.1.3	Is the PT item, a material that meets the definition of "reference material", produced under conditions that meet the relevant requirements of ISO 17034?			
7.1.2	**PT scheme communication**			
7.1.2.1	Does the PT provider make detailed information available about the PT scheme to participants and customers. Does this information include:			
a)	objectives and relevant details of the PT scheme;			
b)	criteria to be met for participation;			
c)	criteria for determining the assigned value and the evaluation of performance;			
d)	confidentiality arrangements;			
e)	critical timelines;			
f)	any fees for participation;			
g)	details of how to apply.			

©SANAS

PART A: TECHNICAL REQUIREMENTS CHECKLIST				
CLAUSE	***ISO/IEC 17043:2023 REQUIREMENTS***	***CAB's COMMENTS*** *The CAB must provide information on **how** requirements have been addressed, documented and/or implemented. Make reference to policies / procedures, incl. clause numbers.*	**C/ NC**	***ASSESSOR's COMMENTS*** *Indicate **WHAT** has been checked and **HOW** requirements have been implemented.*
7.1.2.2	Are the participants and customers advised in a timely manner by the PT provider of any changes in PT scheme design or operation? Yes, in October month introduction of HGI PT was introduced to the market. One month participation, samples were provided for free this is to allow the participants time to decide to include it in their scope of assessment. Also this requirement is stated in ASDP001 management system document section 3.2			
7.1.2.3	Are the records of relevant communications maintained and retained by the PT provider, as appropriate? Yes, sample distribution form			
7.2.2	**Statistical design**			
7.2.2.1	Are the statistical designs developed to meet the objectives of the PT scheme, based on the type of data (quantitative or qualitative, including ordinal and nominal), statistical assumptions, the type of errors and the expected number of results?		C	Document, Ref: ASMF 7.2.2 Statistical design was developed and implemented. Quantitative – all measurands are reported as numerical results
7.2.2.2	Does the PT provider document the statistical design and data analysis methods to be used to determine the assigned value and to evaluate the participant results?		C	The AzureSky coal proficiency testing scheme is designed for quantitative determination of coal characteristics. Each participant receives one bottle of pulverized coal and reports single-determination results for the defined measurands. Statistical design ensures that performance evaluation is objective, consistent, and in line with ISO 13528:2022, Ref: ASMF 7.2.2 It is assumed that participant results are independent and approximately normally distributed after application of robust statistical treatment. Both random and systematic errors may occur among laboratories; the robust method (Algorithm A) minimizes the impact of such variations. statistical assumptions by reviewing data distributions, z-score patterns, and historical control charts from previous
7.2.2.2	Does the PT provider document the reasons for the selection and the assumptions upon which the statistical design and data analysis methods are based?		C	
7.2.2.2	Is the PT provider being able to demonstrate that statistical assumptions are reasonable and that statistical analyses are carried out in accordance with prescribed procedures?		C	

PART A: TECHNICAL REQUIREMENTS CHECKLIST				
CLAUSE	**ISO/IEC 17043:2023 REQUIREMENTS**	***CAB's COMMENTS*** *The CAB must provide information on **how** requirements have been addressed, documented and/or implemented. Make reference to policies / procedures, incl. clause numbers.*	**C/ NC**	***ASSESSOR's COMMENTS*** *Indicate **WHAT** has been checked and **HOW** requirements have been implemented.*
7.2.2.3	In designing a statistical analysis, does the PT provider consider the following:			PT rounds, Ref: ASMF 7.2.2 section 2.
a)	the accuracy, as well as the uncertainty, required or expected for the assigned value for each property or characteristic in the PT scheme?		C	Azuresky Pty Ltd Proficiency Testing Scheme – Hardgrove Grindability Index Report No. 01 October 2025
b)	the minimum number of participants in the PT scheme needed to meet the objectives of the statistical design. In cases where there is an insufficient number of participants to meet these objectives or to produce statistically meaningful analysis of participant results, is the PT provider documenting, and provide to participants, details of the alternative approaches used to assess participant performance?		C	Azuresky applies an internal threshold of n ≥ 10 for the calculation and reporting of z-scores. This criterion is applied to ensure that statistical estimates (mean, standard deviation, and z-scores) are robust and not unduly influenced by small datasets. Where the number of participants is fewer than 10, only indicative results on an air-dry basis are provided.
c)	the relevance of significant figures to the reported participant result, including the number of decimal places?		C	Results are reported to the following decimal places: 1 decimal for Ash, IM, VM; 2 decimals for CV, TS; 0 decimals for HGI, AI
d)	the number of PT items to be measured or tested and the number of repeat measurements or tests to be conducted on each PT item or for each determination?		C	Analysis of Coal: Inherent Moisture (SANS 5925), Volatile Matter (ISO 562), Calorific Value (ISO 1928), Ash (ISO 1171), Total Sulphur (ASTM D4239), Ash Fusion Temperature, Hardgrove Grindability Index (HGI), Ash in Analysis (AI), Ref: ASMF 7.2.2
e)	the procedures used to establish the standard deviation for proficiency assessment or other evaluation criteria?		C	Mean, standard deviation, RSD, Median, Upper and lower control limits, UoM, standard error, min and max, using Anova
f)	the procedures to be used to treat participant results from different measurement or test methods which are not technically equivalent, where permitted by the PT scheme?		C	Document, Ref: ASDP001 section:3.4 is used to guide how results from participants are handled. Participants do not report measurements uncertainties. An effort was made to participants to include them and none

PART A: TECHNICAL REQUIREMENTS CHECKLIST				
CLAUSE	**ISO/IEC 17043:2023 REQUIREMENTS**	**CAB's COMMENTS** *The CAB must provide information on **how** requirements have been addressed, documented and/or implemented. Make reference to policies / procedures, incl. clause numbers.*	**C/ NC**	**ASSESSOR's COMMENTS** *Indicate **WHAT** has been checked and **HOW** requirements have been implemented.*
g)	whether the measurement uncertainty of participant results is being reported and how it will be used to evaluate the participant's performance?		C	of the participants are sending them through See, AS003PT form (this was used at the beginning of operating the scheme ASDP001 document, section 3.4 "Outliers were identified using the criterion of ±3 standard deviations from the mean. Results flagged as outliers were excluded from further statistical evaluation"
h)	the procedures to be used to identify or handle outliers, or both?		C	
i)	where relevant, the procedures for the evaluation of values excluded from statistical analysis?		C	Covered in, ASMF 7.2.2 both objectives and frequency are covered
j)	where appropriate, the objectives to be met for the design and the frequency of PT rounds?		C	
7.2.3	**Determination of assigned values**			
7.2.3.1	Does the PT provider document the procedure for determining the assigned values for the properties or characteristics in a particular PT scheme?		C	ASMF 7.13 assignment of property values and uncertainties was developed and implemented
7.2.3.1	Where applicable, does the procedure take into account the metrological traceability and uncertainty required to demonstrate that the PT scheme is fit for its purpose?		C	Sections 5.3 and 5.8 of document, Ref: ASMF 7.13, procedure take into account the metrological traceability and uncertainty required to demonstrate that the PT scheme is fit for its purpose
7.2.3.2	For PT schemes in the area of calibration, are the assigned values provisioned with metrological traceability?		N/A	

PART A: TECHNICAL REQUIREMENTS CHECKLIST				
CLAUSE	***ISO/IEC 17043:2023 REQUIREMENTS***	***CAB's COMMENTS*** *The CAB must provide information on **how** requirements have been addressed, documented and/or implemented. Make reference to policies / procedures, incl. clause numbers.*	**C/ NC**	***ASSESSOR's COMMENTS*** *Indicate **WHAT** has been checked and **HOW** requirements have been implemented.*
7.2.3.3	For PT schemes in areas other than calibration, was the relevance, need and feasibility for the establishment of metrological traceability and the associated uncertainty of the assigned value determined by taking into account the purpose of the PT scheme?			
7.2.3.4	When a consensus value is used as the assigned value, does the PT provider provide an estimate of the uncertainty of the assigned value as described in the plan for the PT scheme?			
7.2.3.5	Does the PT provider have a policy regarding the disclosure of assigned values?			
7.2.3.5	Does the policy ensure that participants cannot gain advantage from early disclosure?			
7.3.2	**Homogeneity and stability assessment of PT items**			
7.3.2.1	Are the criteria for suitable homogeneity and stability to be established based on the risks that inhomogeneity and instability can impact the evaluation of the performance of participants?		C	Document, Ref: ASDP001 3.5, section 3.5.3 - assessment for stability shall be performed for the PT items to demonstrated sufficient stability and ensure that they will not undergo any significant change throughout the conduct of the PT round
7.3.2.2	Has the PT provider documented the procedures for the assessment of homogeneity and stability?		C	Homogeneity & Stability procedure, Ref: Document no: 7.10 details the process of assessing and monitoring the homogeneity and stability of coal reference materials (RMs) and proficiency testing (PT) samples prepared by the lab.
7.3.2.2	Where applicable, has the PT provider conducted an assessment of homogeneity and stability in accordance with appropriate statistical designs?		C	

©SANAS

PART A:
TECHNICAL REQUIREMENTS CHECKLIST

CLAUSE	ISO/IEC 17043:2023 REQUIREMENTS	CAB's COMMENTS The CAB must provide information on **how** requirements have been addressed, documented and/or implemented. Make reference to policies / procedures, incl. clause numbers.	Cl/ NC	ASSESSOR's COMMENTS Indicate **WHAT** has been checked and **HOW** requirements have been implemented.
7.3.2.3	Is the assessment of homogeneity and stability performed for every PT round after the PT items have been packaged in their final form?		C	**Stability check** 10 samples which were used for the homogeneity assessment were analysed to check for stability, the results are as following: %Dry basis \# Sample / %R1 / %R2 / Sample av (Xt) / Range (Wt) / Range sqd 1 / 13.79 / 14.03 / 13.91 / 0.2371 / 0.0562 2 / 13.97 / 13.77 / 13.87 / 0.1959 / 0.0384 3 / 13.84 / 13.63 / 13.73 / 0.2062 / 0.0425 4 / 13.23 / 13.18 / 13.20 / 0.0515 / 0.0027 5 / 13.75 / 13.74 / 13.75 / 0.0103 / 0.0001 6 / 13.82 / 13.75 / 13.79 / 0.0722 / 0.0052 7 / 13.71 / 13.77 / 13.74 / 0.0619 / 0.0038 8 / 13.68 / 13.59 / 13.63 / 0.0928 / 0.0086 9 / 13.68 / 13.85 / 13.76 / 0.1649 / 0.0272 10 / 13.75 / 13.77 / 13.76 / 0.0206 / 0.0004 Overall Average / 13.72 Standard Deviation / 0.20 SSwithin / 0.784 SSbetween / 0.001 Check value from the homogeneity assessment is 1.135 The mean difference between homogeneity and stability assessments 0.114 Since 0.114<1.135, the PT samples meet stability requirements Yes, refer to ASDP001 3.5.5 "PT items from previous PT rounds, retained for another PT round, the property values or characteristics to be determined in the PT scheme shall be confirmed prior to distribution."
7.3.2.4	Where experimental evidence is needed to assess homogeneity or stability of the PT item (or both), does the PT provider use appropriate methods to assess the homogeneity and stability of the PT item?		C	
7.3.2.5	Have the PT items demonstrated sufficient stability to ensure that they will not undergo any significant change throughout the conduct of the PT round, including storage and transport?		C	
7.3.2.5	When this is not possible, has the stability been quantified and considered as an additional component of the uncertainty associated with the assigned value of the PT item and/or taken into account in the evaluation criteria?		C	
7.3.2.6	When PT items from previous PT rounds are retained for another PT round, are the property values or characteristics to be determined in the PT scheme confirmed again by the PT provider prior to distribution?		C	
7.3.3	**Handling and storage of PT items**			

PART A: TECHNICAL REQUIREMENTS CHECKLIST				
CLAUSE	***ISO/IEC 17043:2023 REQUIREMENTS***	***CAB's COMMENTS*** *The CAB must provide information on **how** requirements have been addressed, documented and/or implemented. Make reference to policies / procedures, incl. clause numbers.*	**C/ NC**	***ASSESSOR's COMMENTS*** *Indicate **WHAT** has been checked and **HOW** requirements have been implemented.*
7.3.3.1	From the time of production to their distribution to participants, does the PT provider ensure that PT items are appropriately identified and stored to prevent contamination, damage or deterioration?		C	Material handling and storage procedure is available to give guidance for transportation, sample handling, retention and disposal of all materials that directly influence the quality and validity of results. Temperature and humidity controlled as necessary (Temperature ≤20 oC, RH 25-45%) in the storeroom Technical forms AS018TF and AS023TF are used to monitor the conditions during testing and in storage room. Other conditions required to be maintained are mentioned in respective test methods. The Business does not handle hazardous items however: AS009IW is a waste management procedure was implemented that aids in managing laboratory waste and how to dispose it.
7.3.3.2	Does the PT provider have appropriate procedures for dispatch to, and receipt from, storage?		C	
7.3.3.3	Are the conditions of stored PT items properly assessed at specified intervals or prior to distribution in order to detect possible deterioration?		C	
7.3.3.4	Where potentially hazardous PT items are used, are facilities available to ensure their safe handling, decontamination and disposal?		C	
7.3.4	**Packaging, labelling and distribution of PT items**			
7.3.4.1	Does the PT provider control packaging and labelling processes to the extent necessary to ensure conformity with relevant national, regional, or international safety and transport requirements?			
7.3.4.2	Does the PT provider document relevant environmental conditions for the transport of PT items?			
7.3.4.2	If necessary, are environmental conditions monitored during transport?		N/A	

PART A: TECHNICAL REQUIREMENTS CHECKLIST				
CLAUSE	***ISO/IEC 17043:2023 REQUIREMENTS***	***CAB's COMMENTS*** *The CAB must provide information on **how** requirements have been addressed, documented and/or implemented. Make reference to policies / procedures, incl. clause numbers.*	**C/ NC**	***ASSESSOR's COMMENTS*** *Indicate **WHAT** has been checked and **HOW** requirements have been implemented.*
7.3.4.3	In PT schemes where participants are required to transport the PT items to other participants, or return them to the PT provider, have the documented instructions for this transport, to ensure the validity of the PT item, been supplied?			
7.3.4.4	Does the PT provider ensure that labels are securely attached to the packaging of individual PT items?			
7.3.4.4	Is the PT provider label designed to remain legible and intact throughout the PT round?			
7.5	Does the PT provider follow a procedure to enable the confirmation of delivery of the PT items?			
7.5	**Control of the PT scheme process**			
7.5.3	**Surveillance of the process**			
	Does the PT provider have a procedure to ensure the validity of the PT scheme? Are surveillance activities planned and reviewed, and the resulting data recorded for the continuous improvement process? **Note:** [see also Section 8.9.2 of the standard - Management review]			ASDP001 section 3.12, state the procedure to conduct he surveillance. ASMF 7.5.3 Surveillance of processes plan was developed and implemented
Additional / General Comments. *This space may also be used by the Assessor to expand on comments in specific sections*				

PART A: TECHNICAL REQUIREMENTS CHECKLIST				
CLAUSE	***ISO/IEC 17043:2023 REQUIREMENTS***	***CAB's COMMENTS*** *The CAB must provide information on **how** requirements have been addressed, documented and/or implemented. Make reference to policies / procedures, incl. clause numbers.*	**C/ NC**	***ASSESSOR's COMMENTS*** *Indicate **WHAT** has been checked and **HOW** requirements have been implemented.*

©SANAS

PART B:
TECHNICAL REQUIREMENTS
VERTICAL CHECKLIST

REQUIREMENTS & COMMENTS.

Compliance = C, Non-compliance = NC. Where a clause is marked as NA, reason must be provided as to why it's not applicable

NB1: *References to ISO/IEC 17043:2023 are in italics. The order of assessment need not follow the order of the checklist. Assessors are expected to know & have the standard, this worksheet is designed as guidance to prompt detailed recording of the process.*

REFER TO ISO/IEC 17043:2023 FOR DETAIL AND FOR CLARIFICATION NOTES.

For Assessor: ***Record at least the number, date and the accredited scope of the PT report(s) selected for Vertical Assessment (s):***

1: ……

2: ……

3: ……

4: ……

Clause	*ISO/IEC 17043:2023 REQUIREMENTS*	*CAB's COMMENTS* *The CAB must provide information on **how** requirements have been addressed, documented and/or implemented. Make reference to policies / procedures, incl. clause numbers.*	*ASSESSOR's COMMENTS* *Indicate **WHAT** has been checked and **HOW** requirements have been implemented.*	C NC NA
7.5.1	**Technical Records** *(state which data and calculations were checked)*			
7.5.1.1	Does the PT provider ensure that technical records for each PT activity contain the results, reports and sufficient information?		The lab has forms that shows that they record and maintain every technical aspect of the laboratory that have direct influence on the quality and validity of results. These can be in a form of equipment verifications, calibration certificates, environmental monitoring and control (temperature & humidity), control charts, temperature recovery.	
	Does the PT provider ensure that technical records for enabling the repetition of the PT activity under conditions as close as possible to the original?			
	Does the technical records include *the date and the identity of personnel responsible* for each PT activity and *for checking data and results?*			

©SANAS

Clause	ISO/IEC 17043:2023 REQUIREMENTS	CAB's COMMENTS *The CAB must provide information on **how** requirements have been addressed, documented and/or implemented. Make reference to policies / procedures, incl. clause numbers.*	ASSESSOR's COMMENTS *Indicate **WHAT** has been checked and **HOW** requirements have been implemented.*	C NC NA
7.5.1.2	Are all the data used to verify the PT items, instructions to participants, the original responses of participants and any other information *included in reports, recorded at the time they are made and identifiable with the specific task?*			
7.5.1.3	Does the PT provider ensure that *amendments to technical records can be tracked to previous versions or to original information* submitted by participants?			
	Are the original and amended data and files *retained,* including the date of alteration, an indication of the altered aspects and the personnel responsible for the alterations?			
6.2	**Personnel** *Check training records of personnel involved in the provision of the PT scheme*			
6.2.1	Does the PT provider have access to a sufficient number of competent personnel to perform its PT activities?			
6.2.2 a)	Does the PT provider ensure that personnel have the competence to:			
b)	perform PT activities for which they are responsible?			
	evaluate the significance of deviations?			
6.2.3	Does the PT provider have a process for managing competence of its personnel?			
6.2.4	Do all personnel of the PT provider (either internal or external) that could influence the PT activities act impartially?			
6.2.5	Does the PT provider have documented information demonstrating competence of its personnel~~,~~ that can influence the results of its PT activities?			
6.2.5	Does the documented information include requirements for education, qualification, training, technical knowledge, skills and experience?			
6.2.6	Does the PT provider authorize personnel to perform specific activities within PT schemes, including but not limited to the following:			
a)	plan PT schemes?			

©SANAS

Clause	ISO/IEC 17043:2023 REQUIREMENTS	CAB's COMMENTS *The CAB must provide information on **how** requirements have been addressed, documented and/or implemented. Make reference to policies / procedures, incl. clause numbers.*	ASSESSOR's COMMENTS *Indicate **WHAT** has been checked and **HOW** requirements have been implemented.*	C NC NA
b)	assess data/information to determine stability and homogeneity, if applicable, as well as assigned values and associated uncertainties of the properties or characteristics of the PT item?			
c)	evaluate the performance of PT participants?			
d)	give opinions and interpretations as well as advice to the participants?			
e)	review and authorize PT reports?			
6.2.7	Does the PT provider management communicate to all personnel their duties, responsibilities and authorities?			
7.2	**Design and planning of a PT scheme**			
7.2.1	**General**			
7.2.1.1	Does the PT provider identify, design and plan those activities which directly affect the validity of the PT scheme and ensure that activities are carried out in accordance with prescribed procedures?		AzureSky Pty Ltd Proficiency Testing Scheme – Participant Instructions AzureSky PT Scheme- Round 1 Date: 13 October 2025, Sample 10/2025 HGI	C
7.2.1.2	When the PT provider intends to introduce significant changes to activities which can affect the validity of the PT scheme, does the PT provider identify and manage the risk to ensure the validity of the PT scheme is maintained?		Risk assessment is a requirement to be performed refer to ASDP001 section 4.3 Implementation to be verified during Initial Assessment	C
7.2.1.3	Does the PT provider develop a documented plan before commencement of the PT scheme that addresses the objectives, purpose and basic design of the PT scheme?		The PT Plan document, Ref: ASMF 7.2.1.3 PT plan Rev 01, Refer to Design and planning of the PT scheme version 00 (this plan has minimal requirements of this clause) A revised 01 Proficiency testing plan was developed and implemented during the running of the scheme (this document addresses all listed points) Implementation to be verified during Initial Assessment	C
	Does the plan include the following information and, where appropriate, reasons for the selection or exclusion of the specific information:			C
a)	the personnel involved in the design and operation of the PT scheme?			C
b)	the activities to be undertaken by external providers of products and services and their contact details?			
c)	criteria to be met for participation in the PT scheme?			
d)	the number and type of expected participants in the PT scheme?			
e)	description of activities to be performed and results to be reported by participants?		Ref: ASMF 7.2.1.3 PT plan Rev 01	C
f)	a description of the range of values or characteristics, or both, to be expected for the PT items?		Ref: ASMF 7.2.1.3 PT plan Rev 01	C

Clause	ISO/IEC 17043:2023 REQUIREMENTS	CAB's COMMENTS *The CAB must provide information on **how** requirements have been addressed, documented and/or implemented. Make reference to policies / procedures, incl. clause numbers.*	ASSESSOR's COMMENTS *Indicate **WHAT** has been checked and **HOW** requirements have been implemented.*	C NC NA
g)	the potential major sources of errors involved in the area of PT offered?			
h)	the requirements for the production, quality control, storage and distribution of PT items?			
i)	arrangements to prevent collusion between participants or falsification of results and procedures to be employed if collusion or falsification of results is suspected?			
j)	a description of the information which will be supplied to participants and the time schedule for the various phases of the PT scheme?			
k)	for continuous PT schemes, the frequency or dates upon which PT items will be distributed to participants, the deadlines for the return of results by participants and, where appropriate, the dates on which measurements or tests will be carried out by participants?			
l)	any information on methods or procedures which participants must use to store, handle, prepare, ship or dispose of the PT item and perform the measurements or tests?			
m)	procedures for the measurement or test methods to be used for the homogeneity and stability testing of PT items and, where applicable, to determine their biological viability?			
n)	preparation of any standardized reporting formats to be used by participants?			
o)	detailed description of the statistical analysis to be used?			
p)	the origin, metrological traceability and uncertainty of any assigned values?			
q)	the treatment of results from different measurement or test methods, where permitted by the PT scheme?			
r)	criteria for the evaluation of the performance of participants?			
s)	a description of the data, interim reports or information to be returned to participants?			

©SANAS

Clause	ISO/IEC 17043:2023 REQUIREMENTS	CAB's COMMENTS *The CAB must provide information on **how** requirements have been addressed, documented and/or implemented. Make reference to policies / procedures, incl. clause numbers.*	ASSESSOR's COMMENTS *Indicate **WHAT** has been checked and **HOW** requirements have been implemented.*	C NC NA
t)	a description of the extent to which participant results, and the conclusions that will be based on the outcome of the PT scheme, will be made public or shared?			
u)	actions to be taken in the case of lost, delayed or damaged PT items?			
7.3	**Production and distribution of PT items**			
7.3.1.1	Does the PT provider establish and implement procedures to ensure that PT items are produced in accordance with the plan and are fit for the PT scheme's purpose?		Yes, for every PT item sufficient homogeneity and stability checks are done and obtained to qualify that they are fit for purpose. Refer to ASDP001 section 3.5 Implementation to be verified during Initial Assessment	C
7.3.1.2	Does the PT provider establish and implement procedures to ensure appropriate selection, acquisition, collection, identification, preparation, handling, storage and, where required, disposal of all PT items?		Yes, procedures from sample prep until storage and distribution were developed and implemented however the disposal of PT items the procedure is not developed because PT items will be recirculated as per ASDP001 3.5.5 point.	C
7.3.1.3	In PT schemes that require participants to sample, prepare or manipulate the PT item and submit it to the PT provider, does the PT provider issue appropriate instructions for preparation, environmental conditions (where applicable), packaging, handling, storage and shipping of the PT item?			N/A
7.3.2	**Homogeneity and stability assessment of PT items**			
7.3.2.1	Do the criteria for suitable homogeneity and stability to be established and to be based on the risks that inhomogeneity and instability can impact the evaluation of the performance of participants?		Yes, this is covered in ASDP001 section 3.5.3 "assessment for stability shall be performed for the PT items to demonstrated sufficient stability and ensure that they will not undergo any significant change throughout the conduct of the PT round"	C

Clause	ISO/IEC 17043:2023 REQUIREMENTS	CAB's COMMENTS *The CAB must provide information on **how** requirements have been addressed, documented and/or implemented. Make reference to policies / procedures, incl. clause numbers.*	ASSESSOR's COMMENTS *Indicate **WHAT** has been checked and **HOW** requirements have been implemented.*	C NC NA
7.3.2.2	Have the PT laboratory documented the procedures for the assessment of homogeneity and stability?		Yes, covered ASDP001 3.5	C
	Where applicable, have the PT provider conducted assessment of homogeneity and stability in accordance with appropriate statistical designs?		Covered ASDP001 3.5 Implementation to be verified during Initial Assessment	
7.3.2.3	Is the assessment of homogeneity and stability performed for every PT round after the PT items have been packaged in their final form?		Covered ASDP001 3.5 Implementation to be verified during Initial Assessment	N/A
7.3.2.4	Where experimental evidence is needed to assess homogeneity or stability of the PT item (or both), does the PT provider use appropriate methods to assess the homogeneity and stability of the PT item?		Covered ASDP001 3.5 Implementation to be verified during Initial Assessment	C
7.3.2.5	Have the PT items demonstrated to be sufficiently stable to ensure that they will not undergo any significant change throughout the conduct of the PT round, including storage and transport?		This is the criteria set to ensure that we mitigate against the risk of instability and inhomogeneity to limit the impact to participants refer to ASDP001 3.5	C
	When this is not possible, have the stability been quantified and considered as an additional component of the uncertainty associated with the assigned value of the PT item and/or taken into account in the evaluation criteria?		It is taken into account in the evaluation criteria.	C
7.3.2.6	When PT items from previous PT rounds are retained for another PT round, are the property values or characteristics to be determined in the PT scheme be confirmed again by the PT provider prior to distribution?		Yes, covered ASDP001 3.5.5 "PT items from previous PT rounds, retained for another PT round, the property values or characteristics to be determined in the PT scheme shall be confirmed prior to distribution." Implementation to be verified during Initial Assessment	C
6.1.2/ 6.3	**Equipment, facilities and environmental conditions**			
6.1.2	Do the measurements or tests conducted under the responsibility of the PT provider, related to PT item characterization or for assessing homogeneity and stability, conducted in accordance with the relevant requirements of ISO/IEC 17025?		Ensures results are reviewed, recorded, and archived in compliance with ISO 17034 record control requirements. The Lab has included environmental requirements needed to perform the method of interest together with its tolerance ranges and monitoring frequencies. (e.g. 19-26 °C for the analysis room, 16-20 °C for the storage facility,	C
6.3.1	Does the PT provider ensure that there are appropriate facilities for the operation of the PT scheme?			C

©SANAS

Clause	ISO/IEC 17043:2023 REQUIREMENTS	CAB's COMMENTS The CAB must provide information on **how** requirements have been addressed, documented and/or implemented. Make reference to policies / procedures, incl. clause numbers.	ASSESSOR's COMMENTS Indicate **WHAT** has been checked and **HOW** requirements have been implemented.	C NC NA
	Do the facilities adversely affect the validity of the results? :		25-45 % RH on both), tolerances and monitoring frequency. Muffle Furnace capable of maintaining 815 ± 10 °C Analytical balance (calibrated, SANAS-traceable) Pulverizer and homogenizer (checked for cleanliness before use, efficiencies performed) Clean sampling scoops and sample trays Labeled 150 g sample bottles with seals Environmental monitoring device (temperature and humidity) Document, Ref ASMF 7.2.3 Environmental is documented to detail hoe the Lab environmental conditions that can influence the validity of the PT items and any measurements or tests carried out Implementation to be verified during Initial Assessment 1. Pulverize the coal material to the required fineness (-212µm). 2. Homogenize the material in a mechanical homogenizer for 4 hours. 3. Extract at random, ten 600g portions of the material (~80 kg batch). 4. Using the flat and heap method, extract 150 g aliquots from ten randomly selected portions and label them with their respective LIMS numbers. Environmental control section 7 states that "Inform supervisor and stop critical activities Implementation to be verified during Initial Assessment Checklist AS018TF and AS023TF ensures adherence to monitoring requirements	C
6.3.2	Does the PT provider ensure that the environmental conditions do not compromise the PT activities, including operations that are undertaken at sites away from the PT provider's permanent facilities or that are undertaken by external service providers?			C
6.3.3	Does the PT provider document environmental conditions that can influence the validity of the PT items and any measurements or tests carried out?			C
	Does the PT provider consider environmental conditions that are required by relevant specifications and measurement or test methods?			C
	Does the PT provider control, monitor and periodically review these conditions and record all relevant monitoring activities?			C
	Are the PT activities be halted if environmental conditions compromise the validity of PT activities?			C
6.3.4	Does the PT provider ensure access control to, and use of, areas affecting the PT activities are managed? Does the PT provider determine the extent of access control based on its particular circumstances?		The Lab commented that access control is not necessary for method code of practice, which is not meeting the requirements of clause 6.3.4 of ISO/IEC 17043	NC
6.3.5	Does the PT provider ensure that there is appropriate separation between neighbouring areas in which there are incompatible PT activities?		Sample preparation is separate from analysis rooms. The analysis of inherent moisture is separated from	C

©SANAS

Clause	ISO/IEC 17043:2023 REQUIREMENTS	CAB's COMMENTS *The CAB must provide information on **how** requirements have been addressed, documented and/or implemented. Make reference to policies / procedures, incl. clause numbers.*	ASSESSOR's COMMENTS *Indicate **WHAT** has been checked and **HOW** requirements have been implemented.*	C NC NA
			furnace (Ash furnace and VM furnace) Implementation to be verified during Initial Assessment	C
	Are actions taken to prevent cross-contamination, interference or adverse influences on PT activities?		Improvement Action request recorded where samples were mixed-up, corrective actions will be evaluated for effectiveness during internal audit Implementation to be verified during Initial Assessment	C
7.3.5	**Instructions for Participants**			
7.3.5.1	Does the PT provider give participants sufficient notice before sending PT items		The schedule/frequency of participation is communicated and agreed during on-boarding i.e. Implementation to be verified during Initial Assessment	C
	Unless the design of the PT scheme makes it inappropriate to do so, does the PT provider provide the date on which the PT items are likely to arrive or to be dispatched?			
7.3.5.2	Does the PT provider give detailed documented instructions to all participants?		Application form contain this information.	C
7.3.5.2	Does the instruction to participants include:			C
a)	the necessity to treat PT items in the same manner as routine samples, including use of routine measurement or test methods, unless there are particular requirements of the PT scheme which require departure from this principle?		The instruction form, informs of the conditions to handle the PT sample and dates to submit results. Ref: Document AS001PT provides that.	C
b)	details of factors which can influence the measurements or tests of the PT items, e.g. the nature of the PT items, conditions of storage, whether the PT scheme is limited to selected measurement or test methods and the timing of the measurements or tests?		Section 4 of the instruction form AS001PT, states the conditions necessary and required for performing coal analysis and this is as per routine test methods Implementation to be verified during Initial Assessment	

Clause	ISO/IEC 17043:2023 REQUIREMENTS	CAB's COMMENTS *The CAB must provide information on **how** requirements have been addressed, documented and/or implemented. Make reference to policies / procedures, incl. clause numbers.*	ASSESSOR's COMMENTS *Indicate **WHAT** has been checked and **HOW** requirements have been implemented.*	C NC NA
c)	instructions for preparing or conditioning, or both, of the PT items before conducting the measurements or tests that would not be considered part of a laboratory's usual expected practices, unless these activities are part of the PT scheme?		All necessary conditions are covered in AS001PT form. Also section 12 of the form, informs of which codes of practice to use for the specific test	
d)	any appropriate instructions on handling the PT items, including any safety requirements?			
e)	any specific environmental conditions for the participant to conduct measurements or tests, or both, and, if relevant, any requirement for the participants to report relevant environmental conditions during the time of the measurement or test?		Implementation to be verified during Initial Assessment	
f)	specific and detailed instructions on the manner of recording and reporting results and associated measurement uncertainties, i.e. when the instructions include reporting of the expanded measurement uncertainty, the reported uncertainty shall include the coverage factor and the coverage probability?		Reporting of results is done through Microsoft form links with specified dates to record results, it gets emailed to participants and it informs of the expiry date to record results. This is covered in section 3.9 of ASDP001 document. In the beginning of the operation of the scheme, participants were required to send through their uncertainties as well, and none of them provided and they are not used in the assignment of values. At the scheme only stated specific test method with its respective code of practice. This is as per design of the scheme until it gets reviewed.	
g)	specific instructions on providing details concerning the measurement or test method used by the participant, where a single specific measurement or test method is not required?		Implementation to be verified during Initial Assessment	
h)	instructions on return or forwarding of the PT items, when applicable?		Covered on ASMF 7.2.1.3 Proficiency Testing Plan	

Clause	ISO/IEC 17043:2023 REQUIREMENTS	CAB's COMMENTS *The CAB must provide information on **how** requirements have been addressed, documented and/or implemented. Make reference to policies / procedures, incl. clause numbers.*	ASSESSOR's COMMENTS *Indicate **WHAT** has been checked and **HOW** requirements have been implemented.*	C NC NA
i)	the last date for the PT provider to receive the results from the participants?		section Actions for lost, delayed, or damaged items This is communicated on Microsoft link forms, the expiry date and time, Section 12 of AS001PT instruction form Addressed in ASMF 7.2.1.3 Proficiency Testing Plan section Actions for lost, delayed, or damaged items Communicated on Microsoft link forms, the expiry date and time. Addressed in Section 12 of AS001PT instruction form	C
j)	information on the contact details of the PT provider for enquiries?			C
7.4.1	**Data Analysis Records**			
7.4.1.1	Have the PT provider recorded results received from participants and analysed the results by using appropriate methods?		Microsoft forms links are used to collect and deposit data entry from participants, it then transfers input data into excel workbook. Data analysis shall generate summary statistics and performance statistics and associated information consistent with the statistical design. Refer to ASDP001 section 3.9	C
	Have the PT provider established and implemented procedures to check the validity of data entry, data transfer, statistical analysis, and reporting?		Covered in section 3.9 of ASD0001	C
7.4.1.2	Does data analysis generate summary statistics and performance statistics and associated information consistent with the statistical design of the PT scheme?		Data analysis shall generate summary statistics and performance statistics and associated information consistent with the statistical design. Refer to ASDP001 section 3.9 and ASMF 7.2.2 Statistical design	C
7.4.1.3	Have the PT provider adopted an appropriate statistical approach to minimise the influence of outliers on summary statistics?			C
7.4.1.4	Where the PT scheme allows participants to use different measurement or test methods, does the PT provider have procedures for treatment of results from different measurement or test methods?		ASMF 7.2.2 Statistical design No data are removed before analysis; Algorithm A provides robustness against outliers. However, results identified as $\|z\| \geq 3.0$ are flagged as outliers. Invalid results (e.g., incorrect method, units, or incomplete data) may be excluded and documented with justification.	N/A
7.4.1.5	Does the PT provider have documented criteria and procedures for dealing with measurement or test results that are inappropriate for statistical evaluation?			C

©SANAS

Clause	ISO/IEC 17043:2023 REQUIREMENTS	CAB's COMMENTS *The CAB must provide information on **how** requirements have been addressed, documented and/or implemented. Make reference to policies / procedures, incl. clause numbers.*	ASSESSOR's COMMENTS *Indicate **WHAT** has been checked and **HOW** requirements have been implemented.*	C NC NA
7.4.1.6	Does the PT provider have documented criteria and procedures to identify and manage situations where PT items that have been distributed and the collected data are subsequently found to be unsuitable for performance evaluation?		ASMF 7.2.2 Statistical Design document stated the "A minimum of 10 participants is required for statistically meaningful evaluation." Not applicable, however Currently AS001PT proficiency testing instruction form or Microsoft link form listed testing methods as follows for the participants: - Ash Content (ISO 1171) - Volatile Matter (ISO 562) - Gross Calorific Value (ISO 1928) - Total Sulphur (ASTM D42390) - Total Moisture (ISO 589) - Inherent Moisture (SANS 2925-2007) - Ash Fusion Temperature (SANS 43:2008) Reducing atmosphere only - Particle Size Distribution (ISO 1953)	C
7.4.2	**Evaluation of Performance**			
7.4.2.1	Does the PT provider use valid methods of evaluation which meet the objectives of the PT scheme?		Yes, the criteria for evaluation of performance is based on consensus statistics from the current round or previous round, then a robust estimate of the standard deviation of participant results is the preferred statistic. This approach enables the performance score of z-scores and to set the standard deviation for proficiency assessment to the calculated estimate of the standard deviation.	C
	Have the PT provider documented the methods and including a description of the basis for the evaluation?		This is covered in ASDP001 section 3.10 Yes, this is covered in ASDP001 section 3.10 and refer to the attached PT report for October to determine objective evidence.	C
7.4.2.2	Where applicable for the objectives of the PT scheme, does the PT provider provide expert commentary on the performance of participants with regard to the following:		Covered in ASDP001 section 3.10 and refer to the attached PT report for October to determine objective evidence. Yes, this is covered in ASDP001 section 3.10 and refer to the attached PT report for October to determine objective evidence.	C
a)	overall performance against prior expectations, taking measurement uncertainties into account?			C

Clause	ISO/IEC 17043:2023 REQUIREMENTS	CAB's COMMENTS *The CAB must provide information on **how** requirements have been addressed, documented and/or implemented. Make reference to policies / procedures, incl. clause numbers.*	ASSESSOR's COMMENTS *Indicate **WHAT** has been checked and **HOW** requirements have been implemented.*	C NC NA
b)	variation within and between participants, and comparisons with any previous PT rounds, similar PT schemes, or published data?			C
c)	variation between measurement or test methods?			C
d)	possible sources of error (with reference to outliers or poor performance) and suggestions for improving performance?			C
e)	advice and feedback to participants as part of the continuous improvement procedures of participants?			C
f)	situations where unusual factors make evaluation of results and commentary on performance impossible?			C
g)	any other suggestions, recommendations or general comments?			C
h)	Conclusions?			C
7.4.3	**PT Reports**			
7.4.3.1	Are the PT reports clear, accurate, objective and comprehensive and include data covering the results of all participants, together with an indication of the performance of individual participants?		Azuresky Pty Ltd Proficiency Testing General Analysis Report - Laboratory Code: 67	C
4.4.3.2	Do the reports include the following (where applicable, or valid reasons exist for not including it):			C
a)	the name and contact details of the PT provider?		Yes, this statement is included:	C
b)	identification of person(s) authorizing the report?		Participants are encouraged to submit any comments or	C

©SANAS

Clause	ISO/IEC 17043:2023 REQUIREMENTS	CAB's COMMENTS *The CAB must provide information on **how** requirements have been addressed, documented and/or implemented. Make reference to policies / procedures, incl. clause numbers.*	ASSESSOR's COMMENTS *Indicate **WHAT** has been checked and **HOW** requirements have been implemented.*	C NC NA
c)	an indication of which activities are provided by external providers when they affect the production or characterization of the PT items or the services provided?		queries regarding the results or operation of the scheme to info@azuresky.it.com or nonhlanhla@azuresky.it.com.	C
d)	the date of issue and status (e.g. preliminary, interim, or final) of the report?		No activities are provided by external providers and the report	C
e)	unique identification that all its components are recognized as a portion of a complete report and a clear identification of the end?		Comment the this is N/A, however the PT report has a unique identification. Confirm during Initial Assessment that the practice is maintained as per the requirements of the standard. Report, it included the following statement on letter to participant section Each participant is allocated a random, unique number for the programme to enable total confidentiality of results" and this covered in ASDP001 section 3.11	C
f)	a statement of the extent to which results are confidential?			C
g)	a unique identification of the report and the PT scheme?			C
h)	a clear description of the PT items used, including necessary details of the PT item's production and homogeneity and stability assessment?			C
i)	the results of participants, including the reported measurement uncertainties?			C
j)	procedures used to statistically analyse the data?			C
k)	statistical data and summaries, including assigned values, range of acceptable results and graphical displays?			C
l)	details of the metrological traceability, and uncertainty of any assigned value?			C
m)	procedures used to establish any assigned value and its uncertainty?			C

©SANAS

Clause	ISO/IEC 17043:2023 REQUIREMENTS	CAB's COMMENTS *The CAB must provide information on **how** requirements have been addressed, documented and/or implemented. Make reference to policies / procedures, incl. clause numbers.*	ASSESSOR's COMMENTS *Indicate **WHAT** has been checked and **HOW** requirements have been implemented.*	C NC NA
n)	assigned values, their uncertainties and summary statistics for measurement or test methods used by each group of participants (if different measurement or test methods are used by different groups of participants)?			C
o)	procedures used to establish the standard deviation for proficiency assessment, or other criteria for evaluation?			C
p)	comments on the performance of participants?			C
q)	information about the design and implementation of the PT scheme?		Uploaded report lab ID 19 and the condition is covered in ASDP001 section 3.11	C
r)	advice on the interpretation of the statistical analysis?			C
s)	comments or recommendations based on the outcomes of the PT round?			C
	Simpler reports are possible for routine reports provided the information is available in scheme protocols or periodic reports available to participants.			C
7.4.3.3	Are reports made available to participants within planned timescales? **Note 1:** In sequential PT schemes, e.g. where the turn-around time can be very long, and in PT schemes involving perishable materials, preliminary or anticipated results may be provided before final results are disclosed. **Note 2** Preliminary or anticipated results allow for early investigation of possible errors.		Emails will serve as objective evidence Our date is the 6th of every month.	NA NA
7.4.3.4	Does the PT provider have a policy for the use of reports by participants and customers?			
7.4.3.5	When it is necessary to issue a new or amended report for a PT scheme or PT round, does the PT provider report include the following:		This section is covered in ASDP001 section 3.11 (a until c)	C

©SANAS

Clause	*ISO/IEC 17043:2023 REQUIREMENTS*	***CAB's COMMENTS*** *The CAB must provide information on **how** requirements have been addressed, documented and/or implemented. Make reference to policies / procedures, incl. clause numbers.*	***ASSESSOR's COMMENTS*** *Indicate **WHAT** has been checked and **HOW** requirements have been implemented.*	**C NC NA**
a)	a unique identification?			
b)	a reference to the original report that it replaces or amends?			
c)	identification of the amendment and a statement concerning the reason for the amendment or re-issue?			
7.4.3.6	When the PT provider issues an amended report to a subset of participant(s), does the PT provider do an analysis of the potential impact on the other participants for that PT scheme and/ or PT round to ensure there is no influence on the general performance of the other participants?		Addressed in section is covered in ASDP001 section 3.11	C
7.4.3.7	When the PT provider issues a statement of participation or performance in addition to the PT report, does the PT provider ensure that statement is not misleading?		Addressed in section is covered in ASDP001 section 3.11	C
Additional / General Comments. *This space may also be used by the Assessor to expand on comments in specific sections*				

©SANAS

Clause	ISO/IEC 17043:2023 REQUIREMENTS	CAB's COMMENTS The CAB must provide information on **how** requirements have been addressed, documented and/or implemented. Make reference to policies / procedures, incl. clause numbers.	ASSESSOR's COMMENTS Indicate **WHAT** has been checked and **HOW** requirements have been implemented.	C NC NA

Team Leader's Signature:	Neville Taylor	**Date:**	
Technical Assessor /Expert's Signature:	Phuti Mathopa	**Date:**	March 2026

©SANAS